

CS3503 – Machine Learning PBL Project Review

Solar Flare Prediction & Space Weather Alert System

Forecasting major (M/X-class) flares 24 hours ahead from solar magnetic-field data

Presentation by

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Problem & Objectives

Problem Statement

A major solar flare releases in minutes the energy of billions of tonnes of TNT. The radiation reaches Earth in about eight minutes and disrupts **HF radio and aviation communication, degrades GPS accuracy, damages satellite electronics and can induce currents that destabilise power grids.**

Satellite operators, polar-route airlines, power utilities and space agencies are all affected, and once an eruption is visible it is already too late to act.

The magnetic field of a solar active region, however, **changes measurably in the hours before it erupts.** The question this project asks is whether that change can be learned from historical observations and turned into a usable 24-hour advance warning.

Objective

- **Identify** the magnetic-field signatures that precede major flares, using SDO/HMI SHARP parameters from the SWAN-SF benchmark.
- **Develop** a leakage-safe pipeline that turns a 12-hour observation window into a 24-hour forecast.
- **Apply and compare** Logistic Regression and Random Forest across two feature iterations, selecting on validation data only.
- **Evaluate** on a locked test partition, scored once, using imbalance-aware metrics (TSS, PR-AUC, HSS2) rather than accuracy.

Expected Outcome

A machine-learning system capable of flagging, **24 hours in advance**, which solar active regions are likely to produce a major flare.

Delivered as:

- a reproducible codebase that recomputes every published number with one command;
- an interactive web dashboard showing the forecast, the risk level and the exact reason for each decision;
- an honest account of the error rates, with the failures shown rather than hidden.

Input, Analysis & Insights

Input / Data Sources

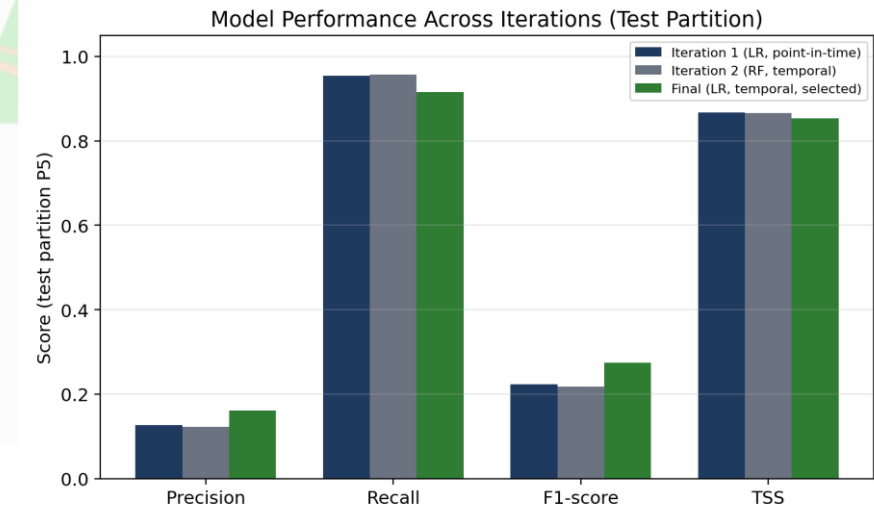
- **SWAN-SF v1.2** benchmark, Harvard Dataverse (DOI 10.7910/DVN/EBCFKM).
- **331,185** multivariate time-series windows spanning **2010-05-01 to 2018-08-17**.
- Each window: **12 h** of SDO/HMI SHARP magnetic parameters at 12-minute cadence — 60 readings × 24 parameters.
- Label $y = 1$ if a **GOES M- or X-class** flare begins within the next 24 h.
- Five chronological partitions: **P1–P3 train (204,559)**, **P4 validation (51,261)**, **P5 test (75,365)**.

Key Insights

- **Trend beats snapshot.** Adding slope and last-value features lifted validation TSS over point-in-time features alone.
- **Accuracy is meaningless here.** Only 1.31 % of test windows precede a major flare, so never alerting already scores 98.69 %.
- A constant-climatology baseline scores **TSS 0.000** and detects nothing; this model scores **TSS 0.853**.
- **Total unsigned current helicity** and **free magnetic energy** dominate the fitted coefficients — consistent with the physics of flare onset.

Analysis / Processing

- 24 parameters × 6 statistics = **144 features** per window (mean, sd, min, max, slope, last value).
- Signed-log transform → median imputation → standardisation — all **fitted on training data only**.
- Partitions are **region-disjoint and time-ordered**: no window and no active region appears in two splits.
- Class imbalance (1.99 % positive in training) handled with `class_weight="balanced"` — a **49:1** weighting.
- Thresholds fixed on validation; the test partition was **scored once**.



Seven candidates were compared; the final model wins on F1 and PR-AUC while holding TSS.

Technical Approach

Architecture

Five stages, end to end:

1. **Ingest & validate** — SWAN-SF archives, timestamp and QUALITY-flag checks.
2. **Window & label** — 12 h observation → 24 h prediction horizon, past data only.
3. **Feature construction** — 144 leakage-safe summary statistics.
4. **Train & select** — fit on P1–P3, choose on P4 by maximum TSS.
5. **Serve** — locked P5 evaluation → risk banding → web dashboard.

Technology Stack

- **Python 3.12** · scikit-learn **1.8.0** · NumPy · pandas · SciPy · joblib · matplotlib
- **Pipeline:** signed-log → median imputer → StandardScaler → Logistic Regression (L2, class-balanced)
- **Dashboard:** HTML / CSS / vanilla JavaScript — the fitted model is re-run in the browser and agrees with scikit-learn to 1e-6
- **Testing:** pytest (202 tests), Playwright headless browser, Ruff + Black
- **Reproducibility:** one command — `python -m solarflare all` — running 375 metric checks and 397 consistency checks

ML Methodology and Implementation Process

Figure 4.1 — Proposed / Implemented End-to-End Solar Flare Forecasting Architecture

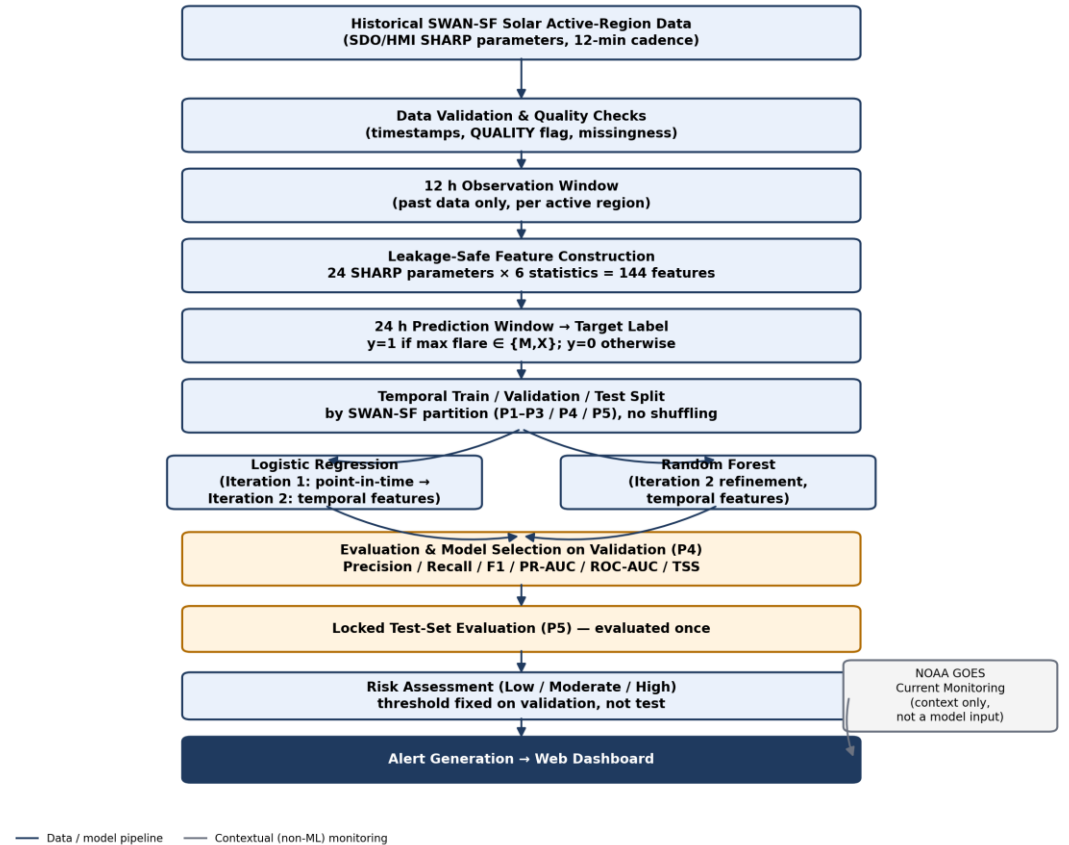


Figure 4.1 — implemented end-to-end forecasting architecture.

Feasibility, Risk and Challenges

Feasibility

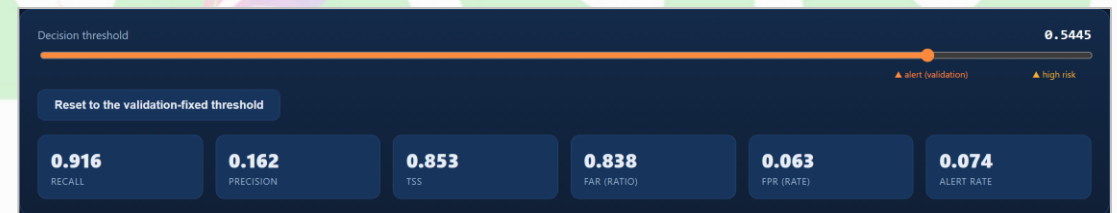
- **Data** — a public, curated benchmark; no licence barrier and no collection cost.
- **Compute** — the full pipeline trains in about 15 minutes on a 4-core laptop. No GPU is required.
- **Deployment** — the model is 144 coefficients, so it runs inside a browser tab with the network switched off.
- **Verification** — every published number recomputes from the repository in about two minutes.

Mitigation

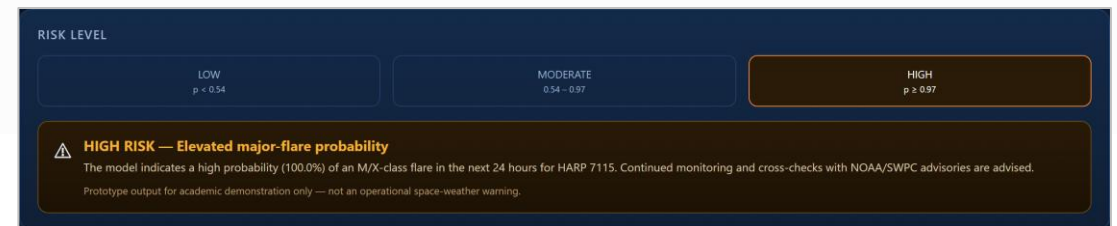
- Class weighting plus **TSS / PR-AUC / HSS2** reporting, with a climatology baseline for context, instead of accuracy.
- Region-disjoint chronological splits, preprocessing fitted on train only, thresholds from validation only — enforced by **21 automated leakage tests**.
- Precision is published rather than hidden, and an operating-point view exposes the trade-off. **78 %** of the false alarms did precede a real B- or C-class flare.
- Archive integrity is verified before extraction; frozen artefacts are committed and write-protected in code.

Risk and Challenges

- **Severe class imbalance.** Positives are 1.3–2.0 % of windows, so accuracy is misleading and naive training learns to answer “no flare” always.
- **Data leakage is the dominant failure mode** in flare forecasting: overlapping windows and shuffled splits inflate published results.
- **Low precision.** At the operating point, **84 %** of alerts are not followed by a major flare.
- **Outputs are not calibrated probabilities** — they rank well but must not be read as literal chances.
- **Two raw partition archives (P2, P5) are corrupt** gzip streams that fail silently mid-stream rather than raising an error.
- **Retraining from scratch is not bit-identical** across operating systems and library versions.



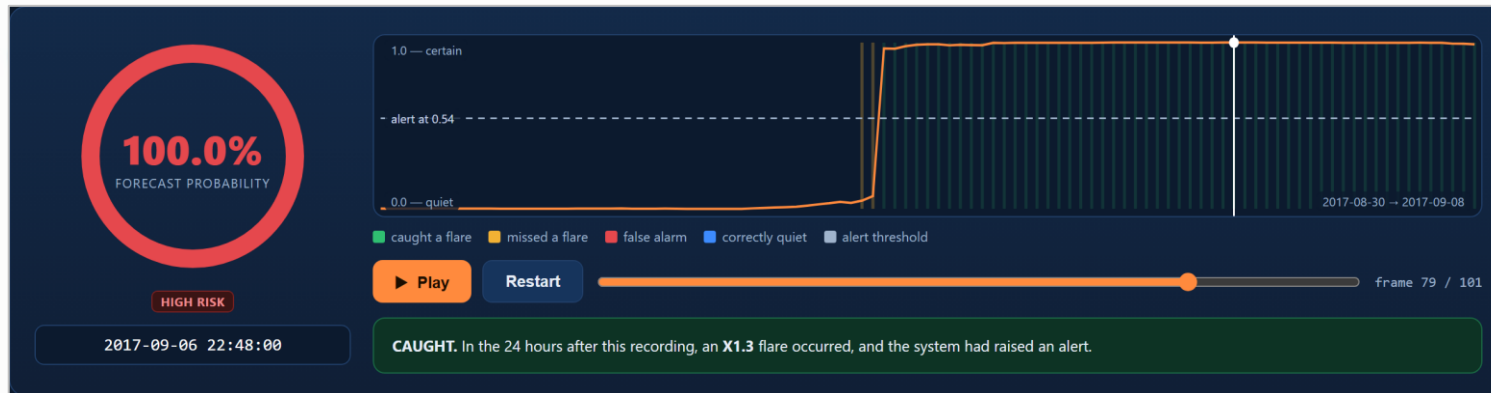
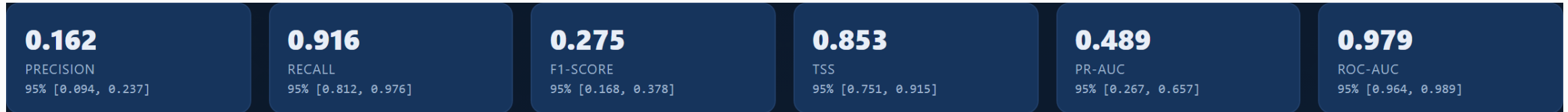
Moving the threshold shows what each choice costs.



An uncalibrated probability is delivered as a three-level band.

Result and Applications

Project Output — interactive dashboard, evaluated on 75,365 unseen windows



Replay of a real active region (HARP 7115, September 2017): the forecast crosses the alert threshold and stays high through the X1.3 flare.

CONFUSION MATRIX (TEST, THRESHOLD FROM VALIDATION)

	Pred: No M/X	Pred: M/X
Actual: No M/X	TN 69,668	FP 4,707
Actual: M/X	FN 83	TP 907

Confusion matrix on the locked test partition.

907 of 990 flare-producing windows were caught (**recall 0.916**, 95 % CI 0.812–0.976), including **19 of 19 X-class windows**, at a false-alarm rate of **0.063**. Precision is **0.162** — shown, not hidden.

Applications

Satellite operations (safe-mode and manoeuvre scheduling) · Polar-route aviation (HF-radio and radiation planning) · Power-grid readiness against geomagnetically induced currents · Launch and spacewalk scheduling · Space-agency situational awareness · A teaching benchmark for severely imbalanced time-series forecasting.

Conclusion, Limitations & Future Scope

Conclusion

We developed a **leakage-safe machine-learning pipeline** using SWAN-SF SHARP magnetic-field parameters to forecast major (M/X-class) solar flares **24 hours in advance**. Seven candidate models were trained on 204,559 windows and compared on a separate validation partition; the selected model — **L2-regularised Logistic Regression on temporal features** — was then scored **once** on 75,365 windows it had never seen.

It achieved a **True Skill Statistic of 0.853** (95 % CI 0.751–0.915), detecting **907 of 990** flare-producing windows — **91.6 % recall**, and **all 19 X-class windows** — at a false-alarm rate of 0.063 and **ROC-AUC 0.979**.

Every published number recomputes from the repository with a single command, and the project ships an interactive dashboard that runs the real trained model in the browser.

Limitations

- **Precision is 0.162** — 84 % of alerts are not followed by an M/X flare (though 78 % do precede a smaller B/C flare).
- Outputs rank well but are **not calibrated** — they are not literal chances.
- **One instrument, one span:** SDO/HMI, 2010–2018. No validation beyond 2018 or on other instruments.
- Only **19 X-class windows** in the test set, so that recall carries a wide interval.
- Two raw archives are corrupt; results rest on the previously extracted features.
- From-scratch retraining is **not bit-reproducible** across platforms.

Future Scope

- **Probability calibration** on a held-out year, so the output can be read as a genuine chance.
- **Sequence models** (LSTM / Transformer) on the raw 60-step series instead of summary statistics.
- Extend to **solar cycle 25** (2019–present) and to other magnetograph instruments.
- Add **magnetogram imagery** alongside the scalar SHARP parameters.
- **Cost-sensitive thresholds** per user — a satellite operator and a power utility do not pay the same price for a miss.
- Ingest a **live SHARP feed** to move from prototype toward an operational service.



Project Video Link

<paste the drive / YouTube URL of the voice-over demo here>

*Thank
you!*



GitHub Repository Link

<https://github.com/Madhavan777/Solar-Flare-Prediction-System-Weather>

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Scan for the repository